Sensor Values → REI Prediction (AI Model)

1. Core Mapping Problem

Sensor Output	What AI Must Infer	Output
MQ-135 ppm	Pesticide class / volatility	REI status
MQ-3 ppm	Bipyridyl / alcohol signature	REI status

2. Model Inputs (Feature Vector)

Feature	Symbol	Range	Source
MQ-135 reading	s■	0–1000 ppm	Sensor
MQ-3 reading	s■	0–1000 ppm	Sensor
Ratio s■/s■	r	0–∞	Computed
Temperature	T	-10 to 50°C	Sensor
Humidity	H	0–100%	Sensor
Time since last baseline	Δt	0–168h	Clock

3. Model Output

Output	Type	Values
Risk class	Categorical	LOW / MED / HIGH / EXTREME
REI estimate	Continuous	0–168 hours (confidence interval)

4. Key Thresholds (Calibrated)

Condition	Inference	REI Estimate
s■ > 200, s■ < 50, r > 4	Group A (organophosphate/chlorinated)	24–48h

s■ > 100, s■ > 150, r < 1	Group B (paraquat/neonicotinoid)	48–168h
s■ < 50, s■ < 50	Baseline / no recent spray	0h (or unknown)
s■ decaying exponentially	Post-application decay	Decay fit → hours since spray

5. Critical Limitation

The model cannot identify specific pesticides. It estimates risk tier and approximate REI range based on volatility patterns, not molecular identification.

6. What You Actually Get

Scenario	Sensor Pattern	AI Output
High s■, low s■, hot/dry	High volatility, Group A likely	"HIGH — assume 24h REI active"
High s■, moderate s■	Group B signature	"EXTREME — assume 48h+ REI active"
Both low, stable	No anomaly	"LOW — no detected application"
Decaying from high	Post-spray decay	"REI likely expiring in X hours" (uncertain)

7. The Honest Output Format

INPUT: [s■, s■, T, H, Δt]
OUTPUT: {
  risk_class: "HIGH",
  estimated_rei_hours: "24-48",
  confidence: 0.6,
  action: "Assume REI active. Do not enter without PPE."
}

No "safe" output. Only "assumed active" or "no anomaly detected."